

MT8002 Elements of Set Theory  
7.3 Ordered Sets. Order Arithmetic

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# Preliminaries

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## Textbook

Derek Goldrei (1996). Classic Set Theory. A Guided Independent Study.

## Convention

The numbers and page numbers assigned to chapters, examples, exercises, figures, quotes, sections and theorems on these slides correspond to the numbers assigned in the textbook.

# Outline

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Introduction

Order Sum

References

# Outline

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Order Sum

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# Introduction

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We shall define two arithmetic operations on linearly ordered sets: the order **sum** and the order **product**.

# Outline

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Introduction

**Order Sum**

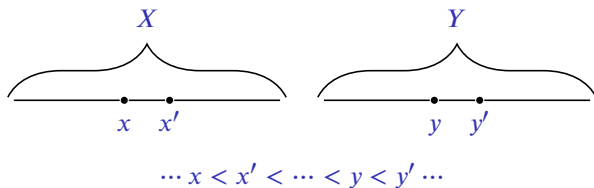
References

# Order Sum

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## Intuitive idea

Let  $(X, <_X)$  and  $(Y, <_Y)$  be two linearly ordered sets. An intuitive idea for adding the sets is the following one.

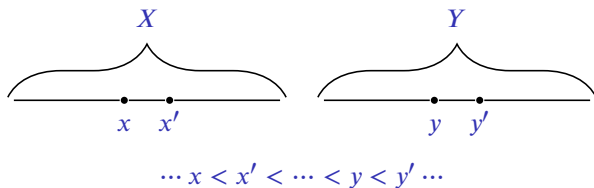


# Order Sum

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## A technical issue

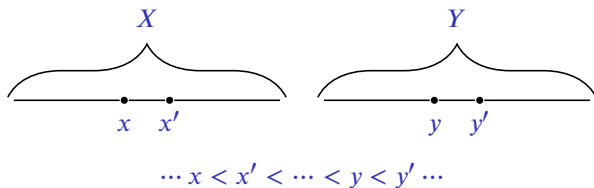
What about if the sets  $X$  and  $Y$  have elements in common?

# Order Sum

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## Intuitive idea

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## A technical issue

What about if the sets  $X$  and  $Y$  have elements in common?

Solution: To use disjoint copies of  $X$  and  $Y$ .

# Order Sum

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## Definition (p. 177)

Let  $(X, <_X)$  and  $(Y, <_Y)$  be two linearly ordered sets. The **order sum** of  $X$  and  $Y$ , denoted  $X + Y$ , is the strictly linearly ordered set

$$(X \times \{0\}) \cup (Y \times \{1\})$$

by the relation  $<$  defined by

$$(x, 0) < (x', 0) \text{ if } x, x' \in X \text{ and } x <_X x';$$

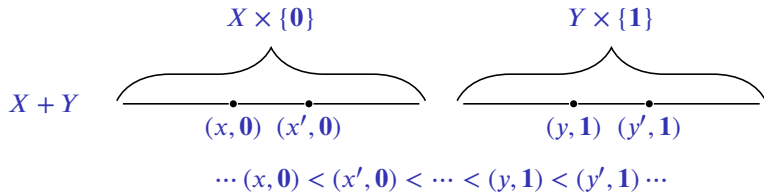
$$(x, 0) < (y, 1) \text{ if } x \in X \text{ and } y \in Y;$$

$$(y, 1) < (y', 1) \text{ if } y, y' \in Y \text{ and } y <_Y y'.$$

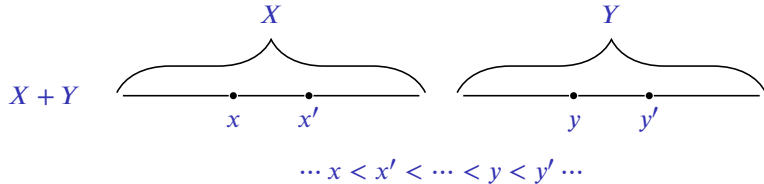
# Order Sum

## Representation

(a) From the definition



(b) From the textbook



# Order Sum

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## Question

Let  $(X, <_X)$  and  $(Y, <_Y)$  be two linearly ordered sets. Is **always**  $(X, <_X) + (Y, <_Y)$  a “new” set? That is,

$$(X, <_X) \not\cong (X, <_X) + (Y, <_Y) \quad \text{and} \quad (Y, <_Y) \not\cong (X, <_X) + (Y, <_Y)$$

# Order Sum

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## Answer

- (a) Finite (non-empty) linearly ordered sets: **Yes, always.**
- (b) Infinite linearly ordered sets: **No, not always.**

# Order Sum

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## Definition (p. 51)

Let  $\mathbf{m}$  and  $\mathbf{n}$  be natural numbers. The (arithmetic) **addition** of  $\mathbf{m}$  and  $\mathbf{n}$ , denoted  $\mathbf{m} + \mathbf{n}$ , is defined by

$$\mathbf{m} + \mathbf{0} = \mathbf{m},$$

$$\mathbf{m} + \mathbf{n}^+ = (\mathbf{m} + \mathbf{n})^+.$$

# Order Sum

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In the following theorem we shall denote the arithmetic addition of natural numbers by  $\oplus$ .

# Order Sum

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## Theorem 7.4

Let  $\mathbf{m}$  and  $\mathbf{n}$  be natural numbers with their usual orders  $\in$ . Then their order-theoretic sum  $\mathbf{m} + \mathbf{n}$  is order-isomorphic to their arithmetic sum  $\mathbf{m} \oplus \mathbf{n}$ .

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# Order Sum

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Proof.

Let  $p(\mathbf{n}) : \mathbf{m} + \mathbf{n} \cong \mathbf{m} \oplus \mathbf{n}$  be a property where  $\mathbf{n}$  is a free variable and  $\mathbf{m}$  is a fixed natural number (parameter).

# Order Sum

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## Proof.

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The proof is by induction on  $\mathbf{n}$ , for fixed  $\mathbf{m}$  (recall that arithmetic addition was defined by recursion in the second argument).

# Order Sum

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(a) **Basis step**  $p(\mathbf{0})$ : Since  $\mathbf{m} + \mathbf{0} \cong \mathbf{m}$  and  $\mathbf{m} \cong \mathbf{m} \oplus \mathbf{0}$ , then  $\mathbf{m} + \mathbf{0} \cong \mathbf{m} \oplus \mathbf{0}$ .

# Order Sum

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- (b) **Inductive step**, for all  $\mathbf{n}$ ,  $p(\mathbf{n}) \rightarrow p(\mathbf{n}^+)$ : The order-isomorphism  $f : \mathbf{m} + \mathbf{n} \longrightarrow \mathbf{m} \oplus \mathbf{n}$  from the inductive hypothesis should be extended to an order-isomorphism from  $\mathbf{m} + \mathbf{n}^+$  to  $\mathbf{m} \oplus \mathbf{n}^+$ , using  $\mathbf{n}^+ \doteq \mathbf{n} \cup \{\mathbf{n}\}$  to send the extra element of each side to the other.



# Order Sum

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Definition (p. 179)

$\omega$  (omega)  $:= (\mathbb{N}, \in)$ .

# Order Sum

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## Example

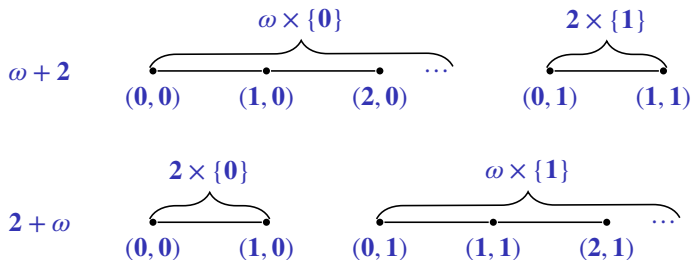
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# Order Sum

## Example

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## Solution



No! In  $\omega + 2$ , the element  $(0,1)$  is a **limit point**, but  $2 + \omega$  has no limit points, so  $\omega + 2 \not\cong 2 + \omega$ .

# Order Sum

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## Remark

From the previous example we know that order sum **is not** commutative respect to order-isomorphism.

# Order Sum

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## Example

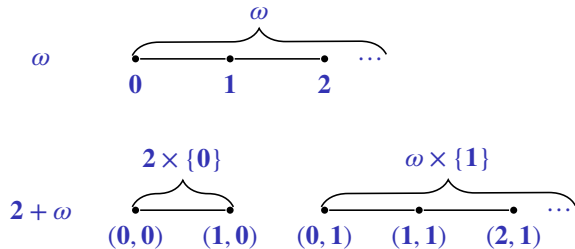
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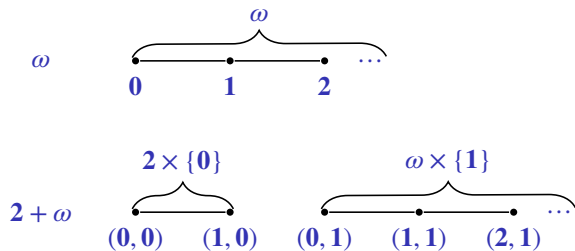


# Order Sum

## Example

Are  $\omega$  and  $2 + \omega$  order-isomorphic? See, Exercise 7.27.

## Solution



Yes! The order-isomorphism is the function defined by

$$f : \omega \longrightarrow 2 + \omega$$

$$i \longmapsto \begin{cases} (i, 0), & \text{if } i \neq 0 \text{ or } 1; \\ (i - 2, 1), & \text{if } i \geq 2. \end{cases}$$

# Order Sum

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## Exercise 7.31

Define a new order  $<'$  on the set  $\mathbb{N}$  so that with this order the set  $\mathbb{N}$  is order-isomorphic to the  $X$  given by:

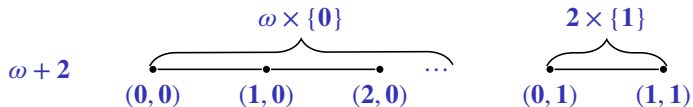
- (a)  $X = \omega + 2$ ,
- (b)  $X = \omega + \omega$ .

(continued on next slide)

# Order Sum

## Solution

For  $X = \omega + 2$ .



On  $\mathbb{N}$  we define  $<'$  so that

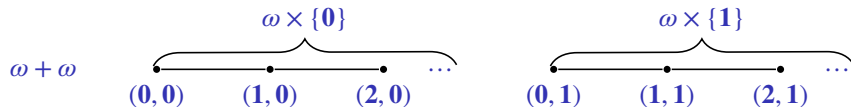
$$2 <' 3 <' 4 <' \dots <' n <' \dots <' 0 <' 1.$$

(continued on next slide)

# Order Sum

## Solution

For  $X = \omega + \omega$ .



On  $\mathbb{N}$  we define  $<'$  so that

$$0 <' 2 <' 4 <' \dots <' 2n <' \dots <' 1 <' 3 <' 5 <' \dots <' 2n+1 <' \dots .$$

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Derek Goldrei (1996). Classic Set Theory. A Guided Independent Study. Chapman & Hall (cit. on p. 2).